

AIM:

To understand the Persistence Layer in Android and implement data storage using SharedPreferences.

Procedure

- 1 Open Android Studio.
 - . Create a new project (Empty Activity, Kotlin).
- 2 Design UI using:
 - . EditText (for input)
- 3
 - . Button (to save data)
 - . TextView (to display stored data)
- 4 Use SharedPreferences to store data as key–value pairs.
 - . Retrieve stored data and display it.
- 5 Run the app and verify persistence after restarting.

Code**activity_main.xml**

```
<?xml version="1.0" encoding="utf-8"?>
<LinearLayout xmlns:android="http://schemas.android.com/apk/res/android"

    android:orientation="vertical"
    android:padding="20dp"
    android:layout_width="match_parent"
    android:layout_height="match_parent">

    <EditText
        android:id="@+id/etName"
        android:hint="Enter Name"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"/>

    <Button
        android:id="@+id/btnSave"
        android:text="Save"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"/>

    <TextView
        android:id="@+id/tvResult"
        android:text="Stored Data:"
        android:textSize="18sp"
        android:paddingTop="20dp"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"/>
</LinearLayout>
```

MainActivity.kt

package com.example.persistencedemo

```
import android.content.Context
import android.os.Bundle
import android.widget.Button
import android.widget.EditText
import android.widget.TextView
import androidx.appcompat.app.AppCompatActivity

class MainActivity : AppCompatActivity() {

    override fun onCreate(savedInstanceState: Bundle?) {
        super.onCreate(savedInstanceState)
        setContentView(R.layout.activity_main)

        val etName = findViewById<EditText>(R.id.etName)
        val btnSave = findViewById<Button>(R.id.btnSave)
        val tvResult = findViewById<TextView>(R.id.tvResult)

        val sharedPref = getSharedPreferences("MyPrefs", Context.MODE_PRIVATE)

        // Load saved data
        val savedName = sharedPref.getString("name", "No data")
        tvResult.text = "Stored Data: $savedName"

        btnSave.setOnClickListener {
            val name = etName.text.toString()

            val editor = sharedPref.edit()
            editor.putString("name", name)
            editor.apply()

            tvResult.text = "Stored Data: $name"
        }
    }
}
```

Result

The application successfully:

- Stores user input using SharedPreferences
- Retrieves stored data after app restart
- Displays persistent data on screen

